

The Living Map — Mine/Tunnel Rescue

IEEE TSYP14 · AESS × RAS Technical Challenge — Phase 1 Technical Report

GitHub: github.com/abderahmenamry-ai/Tsyp14

1. Project Overview

In disaster response, robots exploring collapsed structures, mines, or contaminated zones often operate with no GPS and no communication network. If a robot fails mid-mission, the knowledge it gathered is lost with it. The Living Map addresses this with a two-robot architecture: a **Writer robot** autonomously explores and deposits small radio beacons at points of interest, and a separate **Executor robot** later retrieves that information through a dedicated **Outside Network Area** and continues the mission — without ever needing a direct link to the command post. Our chosen environment is **Mine/Tunnel Rescue**, modeled on a Gafsa-style room-and-pillar mine layout with branching galleries, a dead-end chamber, and a rock-fall-blocked branch. We detect three event types: toxic gas, structural collapse, and trapped workers.

2. Technical Solution & Architecture

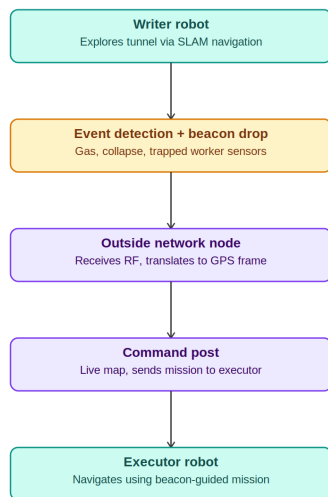


Figure 1 - Living Map system architecture (Mine/Tunnel Rescue)

Fig. 1 – System architecture: Writer → beacon → outside network → command post → Executor.

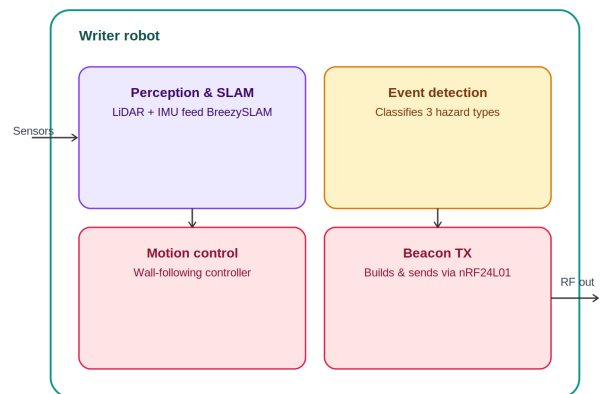


Figure 2 - Writer robot internal architecture

Fig. 2 – Writer robot internal subsystems.

Writer robot autonomy

The Writer combines a simulated 2D LiDAR (360 rays, 8 m range, PyBullet ray casting) and a noisy IMU/odometry model with **BreezySLAM** (RMHC scan matching) to map and localize without GPS. Exploration uses a right-hand-rule wall-following controller with four states (FOLLOW, TURN_LEFT, SEEK, RECOVER). A known SLAM failure mode — heading drift along long, feature-poor corridors ($-0.2^\circ/\text{m}$) — is mitigated with damped odometry/scan-match fusion and a wall-direction heading reference.

Event detection & beacon deposition

Detection triggers when the robot's true position enters one of three predefined hazard zones. Each event is stamped with the SLAM-estimated position (used in the beacon) alongside the ground-truth position (kept for evaluation), letting us quantify localization error end to end.

Beacon message & signal design

Beacons are a compact 16-byte message (ID, event type, priority, local x/y, heading, sensor value, timestamp, TTL) — well within a low-cost nRF24L01's 32-byte payload. Priority derives from event type (trapped worker / collapse = high, gas = medium) and drives an **aging mechanism**: higher-priority beacons start with a longer TTL and rebroadcast more often, so urgent findings persist and are retrieved preferentially.

Frame translation

The Outside Network node converts beacon local coordinates to GPS via a one-time manual calibration at the tunnel entrance (GPS position + compass bearing), combined with a flat-earth approximation valid at

this scale. Manual calibration was chosen over an onboard magnetometer, since compasses are unreliable near a mine's metal structures and mineral content.

Outside network area

All robot-command-post communication passes through the Outside Network node — no direct link exists, satisfying the challenge's architectural constraint. The node receives beacons, translates coordinates, and forwards to the command post, which aggregates data, renders a live map, and builds the Executor's mission as a priority-sorted list of unvisited targets.

Executor robot

The Executor consumes the command post's prioritized list and navigates to each target using an **A* planner** over its own SLAM occupancy map (walls inflated 0.35 m, unknown space passable but costlier), followed by pure-pursuit path tracking with periodic replanning. Across four tested random seeds, the Executor successfully retrieved all 3 beacons.

Results (simulation, default seed)

Writer: ~448 s, ~146 m driven, zero collisions, all 3 events reached. Mean SLAM position error 0.14 m (max 0.28 m) vs. 4–8 m odometry-only drift over the same run. Executor: 3/3 beacons visited across seeds, ~125–150 s, ~51–63 m driven, max position error 0.17–0.26 m.

3. Event Detection Sensors

All three sensors use a plain analog or digital pin — no I2C/SPI — keeping wiring and debugging simple for a beginner team. Each is a low-cost, well-documented breakout module.

Toxic gas — MQ-2. Tin-dioxide semiconductor sensor, responds to LPG, propane, methane, hydrogen, alcohol and smoke over ~300–10,000 ppm. Analog output read via a voltage divider (ESP32 ADC is 3.3V-max). Not selective for toxic gases specifically and sensitive to temperature/humidity — used as a threshold-above-baseline anomaly trigger, not a calibrated measurement.

Structural collapse — SW-520D ball tilt/vibration switch. A sealed rolling-ball switch (no mercury) reporting a binary tilted/not-tilted state, trips at ~14–45° of tilt. No calibration, no library. Firmware debounces and requires repeat triggers in a short window, since it also trips on the robot's own bumps.

Trapped worker — HC-SR501 PIR. Standard hobby PIR, detects a moving warm body across a cone up to ~7 m / 120°. Needs 30–60 s warm-up (ignored in firmware). Digital 3.3V-safe output. Limitation: most reliable while the robot itself is stationary, and cannot detect a worker who is still.

4. Implementation Plan (Phase 2 Roadmap)

Item	Qty	Est. cost	Notes
ESP32 dev board	3	\$15–21	Writer, Executor, outside-network node
Motor driver + DC motors + chassis	2 sets	\$16–25	Differential drive per robot
MQ-2 / SW-520D / HC-SR501	1 each	\$5–9	Event sensors (Writer only)
nRF24L01 radio	3	\$4–8	Writer, Executor, outside node
Battery + regulation + wiring	–	\$13–20	5V motor/sensor rail, 3.3V logic
Base total		≈ \$55–75 — matches Phase 1 budget target	
Navigation sensing (A: ultrasonic+servo / B: 2D LiDAR)	2 sets	\$26–40 / \$146–210	+ IMU each; Option B matches our simulation

Build order: (1) one robot driving → (2) LiDAR/IMU + SLAM on hardware → (3) event sensors, tested individually → (4) radio link, beacon TX/RX test → (5) second robot (same build, no sensors) + goal-seeking controller → (6) outside-network node (simplest build) → (7) full integration test.

Testing plan: bench-test each sensor alone (gas baseline+rise, tilt+debounce, PIR range with robot stopped, motor/IMU straight-line and turn error) → navigation test in a hallway/cardboard maze, target pose error comparable to simulation → radio range/packet-loss test at increasing distance → shared start-frame and calibration checks → full integration test in a mock tunnel with safe hazard stand-ins.

Rough timeline (~8 weeks): wk 1–2 parts + robot #1 driving; wk 3–4 SLAM on hardware + sensor bench tests; wk 5 radio link + range tests; wk 6 robot #2 + outside node; wk 7 integration testing; wk 8 fixes, retest, demo recording. Biggest risks: part shipping delays, real-sensor-noise SLAM tuning, and radio range underground — a spare week is built into wk 8 for these.

5. Failure Cases

SLAM heading drift in long, featureless corridors. Stock BreezySLAM drifts $\sim -0.2^\circ/\text{m}$ in straight corridors (worse while turning: -36° over 18s). A perfectly featureless tunnel gives scan matching nothing to anchor to. *Mitigation:* damped scan-match/odometry/gyro fusion, no correction while rotating fast, and a wall-direction heading reference — reducing mean error to 0.14 m vs. 4–8 m odometry-only. *Remaining:* the wall-direction reference assumes an orthogonal mine layout; irregular tunnels would need it disabled.

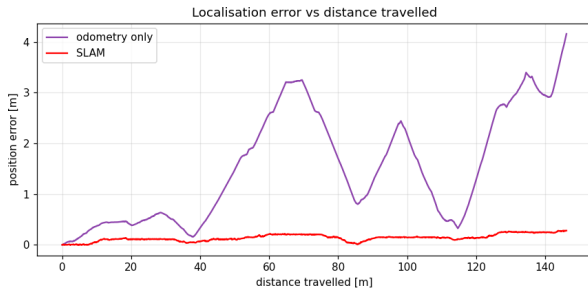


Fig. 3 – SLAM (red) vs. odometry-only (purple) position error.

Sensor false positives/negatives. MQ-2 is non-selective and drifts with temperature/humidity — mitigated via sustained-rise-above-baseline triggering. PIR can false-trigger on warm moving air and misses a motionless worker — a risk the sensor choice cannot fully remove. The tilt switch trips on the robot's own bumps — mitigated with debounce + repeat-trigger windows. Simulation does not exercise these cases; Phase 2 bench tests will.

RF range limits. nRF24L01 range is cut further by rock/metal underground; beacons can be lost if the Writer exceeds range of the outside node. Priority-based TTL/rebroadcast gives urgent beacons more chances through, but does not help once fully out of range. *Known direction (not built):* a multi-node mesh relay, and PA+LNA radio variants if range tests demand it.

Frame translation assumptions. Flat-earth approximation + one-time manual calibration is accurate at single-tunnel-network scale; a 2° heading error shifts a beacon 100m out by $\sim 3.5\text{m}$. Manual calibration was chosen over a magnetometer due to underground magnetic interference. Executor driving is unaffected by GPS calibration error, since it uses beacon coordinates directly in the shared entrance frame.

Executor navigation — tested only in simulation. An earlier straight-line controller reached only 1 of 3 beacons. The current A*-based planner + pure pursuit reaches 3 of 3 across four seeds (max position error 0.17–0.26m). *Remaining limits:* treats unknown space as passable (can head into unexplored dead ends before replanning), steering gains reused from the Writer's wall-follower and not separately tuned, assumes a static tunnel, and all results are simulation-only — real wheel-slip and sensor noise will be worse. Phase 2 repeats this test physically.